

Research article

Intelligent data-driven condition monitoring of power electronics systems using smart edge–cloud framework

Sachin Kumar Bhoi^{a,b}, Sajib Chakraborty^{a,b}, Boud Verbrugge^{a,b}, Stijn Helsen^b, Steven Robyns^b, Mohamed El Baghdadi^{a,b}, Omar Hegazy^{a,b,*}

^a MOBI-EPOWERS Research Group, ETEC Department, Vrije Universiteit Brussel (VUB), Pleinlaan 2, Brussel, 1050, Belgium

^b Flanders Make, Gaston Geenslaan 8, Heverlee, 3001, Belgium



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ABSTRACT

The ongoing revolution in industrial production- Industry 4.0, is driven by transformative technologies such as the Industrial Internet of Things (IIoT), Artificial Intelligence (AI), single-board computers, and 5G communication. As the trend towards IIoT continues, an increasing number of industrial drive systems and their fleets are being connected to the cloud. This enables the manufacturers to perform condition monitoring (CM) and streamlined maintenance activities. At the heart of these drive systems are Power Electronics Systems (PESs), which operate at high switching frequencies (10 kHz–1 MHz) to efficiently transfer electrical power and deliver it to a load in a controlled manner. However, due to their functionalities and the presence of semiconductor switches, PESs are susceptible to failure, necessitating effective condition monitoring (CM) for fault detection and improved lifetime. Link to this issue, to enable CM based on high-frequency data, an industrial site with multiple electric drives is required to record data up to 15TB/week. Therefore, there is a demand from industrial partners to establish intelligent communication between a fleet of physical systems and the cloud to reduce transmission, storage, and bandwidth costs, as well as to enable real-time fault detection and learning from fleet operations. This paper proposes an intelligent edge–cloud computing methodology to address the challenge of high-frequency data monitoring for PESs, focusing on novelty detection and selective data transmission to reduce transmission costs. The methodology involves developing a novel edge–cloud framework that incorporates a neural network-based novelty detector for selective data transmission from physical systems to the cloud. The proposed methodology is evaluated through hardware tests, demonstrating a significant reduction in data transmission (94%) and potential cost savings of up to €5.9k/year for a single remote system. 95.6% detection accuracy of the PQ phase is obtained during experimental tests over 590 samples. Thus, this paper contributes to the vision of the smart grid and IIoT by analyzing the Power Quality (PQ) monitoring problem of a three-phase grid and showcasing the capability of the proposed framework in terms of novelty detection and data transmission cost reduction. To conclude, the proposed intelligent edge–cloud computing methodology offers a promising solution for effective condition monitoring of PESs, with potential cost savings and improved fault detection capabilities. By leveraging advanced technologies and intelligent data-driven approaches, this framework advances the goals of Industry 4.0 and paves the way for efficient and reliable industrial operations in the digital age.

* Correspondence to: Vrije Universiteit Brussel, Belgium.

E-mail address: omar.hegazy@vub.be (O. Hegazy).

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1. Introduction

The emergence of the IIoT, which is one of the key areas of Industry 4.0, together with cyber–physical systems (CPS) and data analytics [1], has provided a pathway for real-time CM, anomaly detection (AD), and prognosis and health management (PHM) in the field of engineering. The backbone of the IIoT is a network of physical systems, nodes or edges that are dispersed globally and transmit data to a centralized computing server referred to as the cloud. The applications of IIoT technology are broad and encompass areas such as automotive, traction systems, aerospace, industrial machines, medical equipment, distributed power generation and smart grid technologies. Most of these applications are of an electrical nature and PESs are utilized to efficiently convert electrical energy between power sources and loads. In recent years, the development of PESs has become increasingly complex, incorporating advanced cooling systems, module-integrated power switches and sensors, real-time digital signal controllers and advanced control techniques. As a result, the deployment of PESs in cutting-edge applications presents specific requirements and distinct challenges for IIoT applications for real-time CM and fulfilling service lifetime targets, including edge node management, data offloading, data bandwidth and latency, the integration of AI at the edge, and data security, as outlined in [2,3]. Furthermore, PESs also present their own set of requirements and challenges, such as the need for high-frequency sampling, limited computational capabilities and long-term mission-critical performance [4].

Two primary methods exist for deploying IIoT for PESs: (a) Cloud-centric approach and (b) Edge-centric approach [2,5]. In a cloud-centric approach, all data are uploaded to the cloud to be analyzed by data-driven models. The main disadvantage of this method is that the amount of data to be sent to the cloud for reliable CM/AD application is too high and economically not feasible. To give an example, if a leading industrial supplier with a connected fleet of 150000 motor drives had to collect high-frequency current, voltage and vibration data and send the data to the cloud, it would cost up to \$38000 per year to transmit 50 GB/day for cloud-based CM activities. In the edge-centric approach, the CM/PHM algorithm runs locally on the edge and a limited amount of data is sent to the cloud. The obvious problem with this approach is the limited storage capability of edge devices to store large historic data sets. Re-training the deployed model requires an expensive graphical processing unit (GPU) which is costly and sharing the knowledge with the entire fleet without sharing the training data is still an advanced research problem [6]. An example of an edge device is the NVIDIA Jetson Nano, which is equipped with a quad-core ARM Cortex-A57, 4 GB of RAM, and 32 GB of storage, and costs €129. The optimal ratio of edge devices to the number of PESs to be monitored depends on the volume of data that needs to be processed in real-time. These types of edge devices are typically designed for deploying pre-trained models, as training on the edge device is not practical [7].

Recent advancements in machine learning (ML), AI, and 5G communication have facilitated edge computing to address the practical challenges associated with IIoT [8,9]. Edge computing can be defined as a novel computing model that distributes computing and storage resources in proximity to the physical systems, located at the edge of the network [10–12]. In addition, edge computing brings enterprise applications closer to data sources such as IIoT devices or local edge servers, as opposed to collecting and processing the data in the cloud. As the number of edge devices increases, the volume of data transfer to the cloud from real-field operations will increase exponentially [5,13]. Recent research studies have emphasized the importance of an efficient, robust and practical edge–cloud computing framework combination for optimal performance of the IIoT technology, with particular attention paid to edge-computing architectures for complex and real-world industrial operations. This would provide better proximity to data, faster insights, improved response times and better bandwidth availability due to data processing and analysis closer to the point where it is created. Besides, as data does not require to traverse over a network to a cloud or data center to be processed, latency will be significantly reduced [2,4,9,13].

Nevertheless, a paradigm shift is ongoing in Industry 4.0 which demands more than just the right artificial intelligence and/or machine learning algorithms, but also appropriate quality measures to achieve an optimal balance between data transmission and monitoring/predicting performance. Therefore, an intelligent data-capturing framework is required to selectively and efficiently record the right data at the right time and right place. Data transmission reduction has been a well-researched topic in the IoT domain. A comprehensive overview of data reduction solutions at the edge is provided in [14] and highlights data filtering and data clustering as the most used techniques. The gap in the current research is the absence of a generalized framework design to integrate data-reduction techniques at the edge, which can be seamlessly integrated into the edge–cloud architecture of IoT solutions.

As already briefly stated before, a key application of edge computing is the practical implementation of smart grids, contributing to the realization of smart meters [15] and globally spread electric vehicle charging stations [16]. Edge intelligence may be regarded as the most realistic strategy for the deployment of services associated with smart grids [17]. Especially for PQ, which has a direct economic impact on industrial consumers, an accurate identification is a prerequisite for the implementation of effective solutions that can mitigate their impact [18,19]. Major causes of PQ problems are non-linear loads [20]. The increase in distributed renewable energy generation, fast charging infrastructure for electric vehicles and usage of PESs in industrial systems are only going to increase PQ issues. To effectively detect PQ events, it is necessary to sample grid voltages at high frequencies and process all data in real-time.

To address the open research topics, viz. scalability, AI and edge hardware, this paper showcases the working of edge and cloud software simultaneously using an intelligent data-capturing framework with data reduction on the edge and application software on the cloud. The paper builds further on the developed-edge software in [21], which provides a real-world demonstration of data reduction using edge data classification, and the results of [22] which tackled the detection of PQ events of a three-phase grid used for industrial motor drives. In this paper, the developed framework is implemented for CM and classification of PQ events of a three-phase grid voltage, contributing to the vision of a smart and sustainable grid [23]. Novelty detection is used to classify incoming PQ data between normal and abnormal data sets, where the abnormal data set is transmitted to the cloud for further

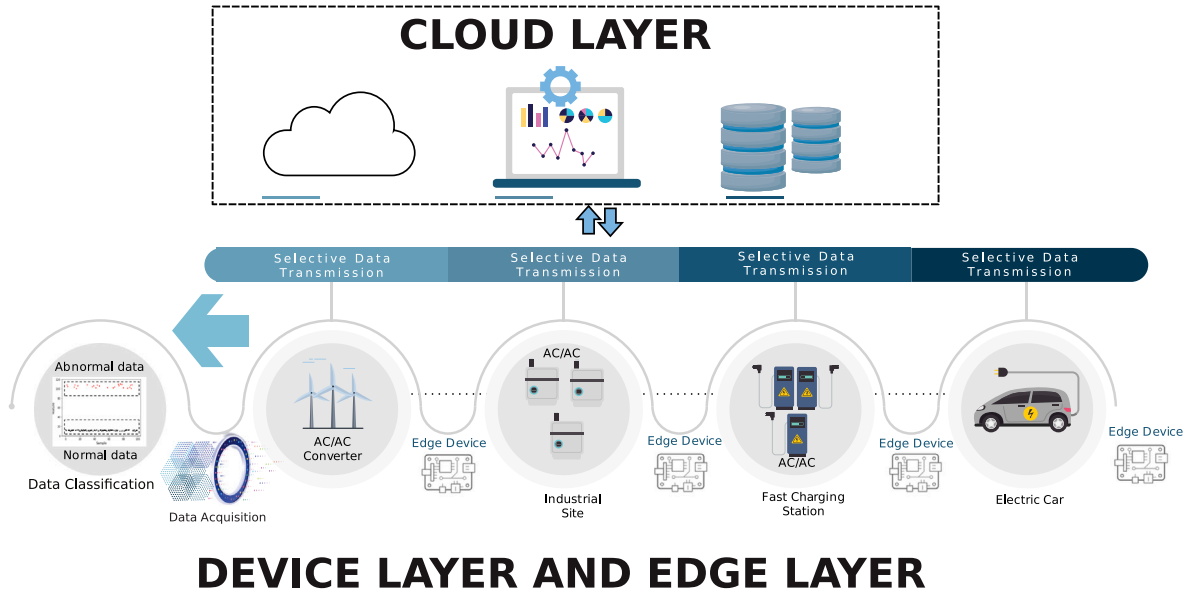


Fig. 1. Intelligent Edge-Cloud computing framework: Overview on usability and applications.

classification by a tree-based ML algorithm. Furthermore, this article outlines the cost savings associated with data transmission by performing a 7-day continuous test on the three-phase grid supply.

The integration of edge computing and cloud functionalities and its potential application in PESs is explored in Section 2. The physical implementation of the framework for monitoring PQ events in three-phase grid voltage is discussed in Section 3 to demonstrate its applicability in smart grid applications. Sections 4 and 5 detail the development of edge and cloud software, respectively. In Section 6, the results of testing the framework with two different test setups are presented to validate its effectiveness. In Section 7, the conclusions of the paper are formulated.

1.1. Current research: contributions

After a comprehensive analysis of the existing literature concerning deployable IIoT architectures for monitoring PESs, several research gaps have been identified. This paper aims to address the following research gaps:

- The research is towards exploring the feasibility of real-life deployment of the IIoT software to industrial setups. Towards this, the implementation is based on a very structured and practical edge-cloud software architecture. The edge software is virtually divided into specific sections, to handle data structure creation, intelligent data processing and classification, and data transmission. All the sections run in parallel processes for real-time data processing and transmission to the cloud. The cloud software shown is, capable of receiving transmitted data from edge, process and modifying it and providing meaningful results.
- The paper contributes to the smart grid vision by tackling the PQ problem in the context of IIoT solutions. An edge-deployable and optimal neural network novelty detector is used to classify between normal and abnormal PQ events on the edge. The edge-cloud architecture allows selective data transmission of only abnormal voltage signals for further analysis on the cloud database.
- The cloud further classifies the dataset using an optimized machine-learning model. Specifically, as a requirement from an industrial partner, the works show the detection of PQ events, specific to the phase in which it has occurred, from available line-to-line voltage information namely “Vab”, “Vbc”, and “Vca”.
- Finally, the developed software is tested with i. A grid simulator (for fault emulation) for validation of the classification algorithms. ii. A three-phase grid supply that powers several motor drives and power electronics testing equipment in the laboratory.

2. Edge-cloud computing framework with intelligent data capturing

The need for a hybrid approach is clearly evident from the current state-of-the-art practices regarding data management in PESs. The proposed edge-cloud-based hybrid methodology is depicted in Fig. 1. Modern systems such as electric vehicles, fast charging stations, industrial production sites, electric aerospace, railways, and distributed power generators (wind and solar) often deploy multiple PE devices. Each of these systems is considered an edge node, which comprises multiple PES from the device layer. Each

of these systems from the edge layer selectively sends data from multiple PESs to the storage and computing server in the cloud layer. The advantages of such a hybrid computing framework is as follows:

1. The selective data transmission by edge devices reduces the data bandwidth, latency and transfer cost without compromising the overall information of the data set present in the cloud. This is achieved by triggering high-frequency raw data transfer on the occurrence of a unique or faulty operating point. This approach allows for improved response time and speed by bringing data processing near the data source.
2. The main functionality of the cloud platform is gathering processed data from AI-powered CM models, which run in edge devices and deliver to cloud AI applications as a secondary phase of processing for long-term inferences. Thus, the computational overhead of the whole architecture is distributed between the edge and cloud node to improve efficiency, speed, latency and cost, avoiding the need to send the high-frequency raw data generated in real-field to the cloud platform.

The function of the edge node is to capture high-frequency data from PESs and to process it in real-time to perform CM/AD activities. The edge node then has the following functionalities:

- **Novelty and fault detection:** The difference between normal and faulty operating points is captured through ML models on edge. These models are trained in a supervised or unsupervised manner based on the availability of training data and expertise.
- **Selective data transmission:** To reduce the cost of data transmission to the cloud, faulty data and unique normal operating point data are transferred to the cloud. In this manner, the cloud will have access to a very compact and full operating data set of the PES.
- **CM and PHM services:** The condition of the PES can be monitored through various data-driven and mathematical models. These models can act as virtual sensors or abnormality classifiers utilizing real-time data to monitor the PES. Data from multiple edge sites can help to further extend these models to perform preventive PES maintenance. An example is to deploy models to estimate the junction temperature of the power devices (MOSFET/IGBTs) on the edge device and transmit the calculated temperature to the cloud.

Multiple edge nodes will selectively transmit data to the cloud and receive health management updates. The cloud receives data from a fleet of edge devices. AI and big data analysis are deployed on the cloud to analyze fleet data and detect abnormal operations. Cloud functionalities include

- **Digital twin platform:** Iteratively updated multi-physics-based high-fidelity virtual prototypes can be created by leveraging real-time data from edge devices to develop digital twins of PESs. These digital twins incorporate models for lifetime estimation and reliability analysis, which enable real-time monitoring of the health of PESs. These trends assist in the development of advanced and integrated PES designs with a focus on reliability.
- **Fleet management:** A fleet of PESs can be managed in real-time for health degradation features over a lifetime, for comparing health features between systems in the fleet, and for assessing fault severity between systems in the fleet.
- **Data visualization and user interface:** The data of the fleet is visualized for human interaction. High-level human decision-making is indispensable for the highest performance output.

2.1. Functional description

The functional block diagram of the proposed hybrid cloud–edge framework is given in Fig. 2. The concept of snapshot structure, intelligent trigger and overall edge software framework has already been proposed and detailed in [21]. The physical system of a PES comprises a power electronics converter module with electrical I/Os, I/O sensors and a data acquisition system (DAQ). The DAQ is critical to the proposed framework. Hardware implementation of the DAQ presents a challenge for the PES. High-frequency sampling (i.e., from a few kHz to MHz), high voltage isolation and cost-to-quality ratio optimization are a few of the critical areas of commercial development. This section summarizes each functional sub-section of the hybrid cloud–edge framework.

2.1.1. Snapshot creator

The data from PESs mostly contain high-frequency noise from dv/dt and di/dt of the semiconductor power switches. The continuous data from the DAQ is pre-processed, filtered and converted to useful real-time data. The continuous stream of data has to be divided into batches of data. Further computations, such as feature extractions and health indicators, are derived from these data batches. A snapshot is a custom data class, which stores the data batch. A brief description of the components of a snapshot is as follows.

1. **Timestamp:** Time of creation of a snapshot in the edge device
2. **Metadata:** Additional information on the data such as PES and DAQ information, machine identifiers and configurations.
3. **Signal:** Signal is a data class containing the raw data, the sampling rate and the signal name. Additional information can be added in the form of metadata for a particular signal.
4. **Features:** Features are calculated health indicators or mathematical parameters derived from the snapshot raw data. The signals are passed through the calculation algorithm and results are added to the snapshot.

The overall data structure of a snapshot is summarized in Fig. 3. From an application-specific point, two variables need to be optimized, 1. the sampling frequency of the signal, 2. the time length of the snapshot. The computational requirement and performance depend on these two variables.

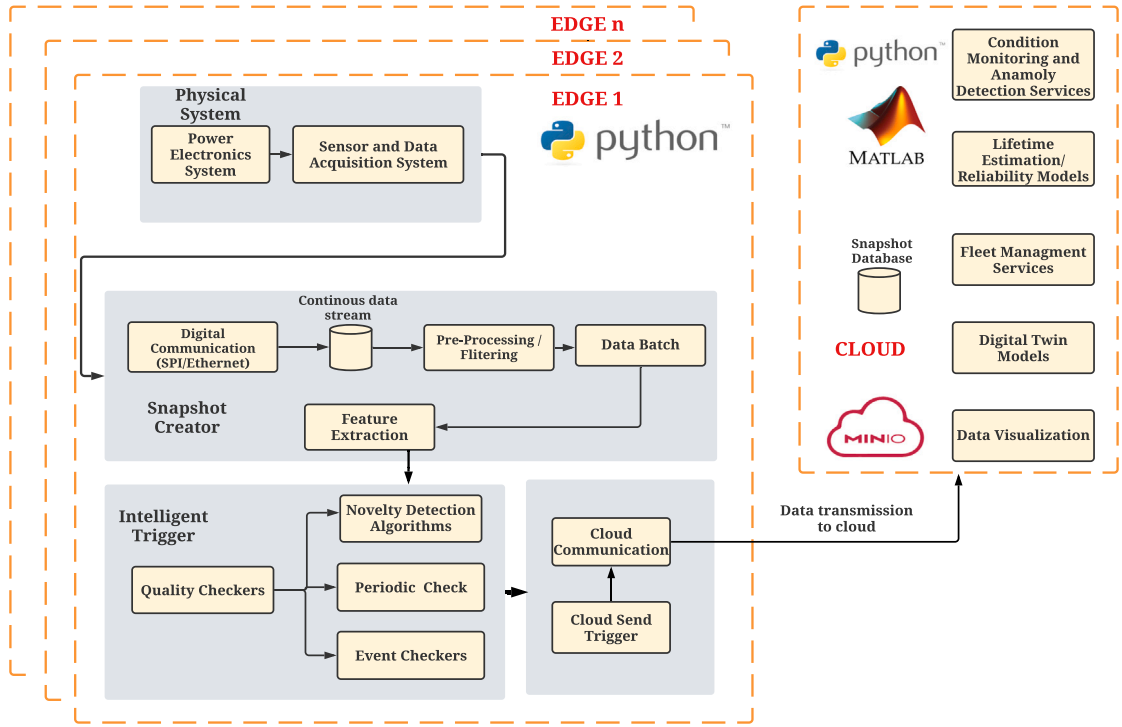


Fig. 2. Stepwise functional detail of the AI-powered computing framework, including both edge node and cloud platform.

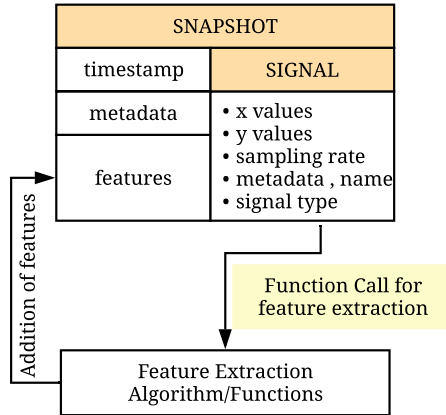


Fig. 3. Structure of a typical snapshot.

2.1.2. Intelligent trigger

An intelligent trigger uses the extracted features of the snapshot to decide which data has to be transmitted to the cloud layer. A scenario would be transmitting calculated features for all the data and signals and features for faulty system operating points. The main subsystems of the intelligent trigger are

- **Data classifier and novelty detector:** This function contains hybrid models to classify the operating condition of the PES. Any detection of abnormality will trigger a full data transmission.
- **Periodic check:** A transmission trigger is always generated periodically to send data to the cloud. One potential use of this trigger is adjusting it dynamically to provide the cloud with the necessary normal operating point data. When the edge is first initiated, the cloud requests a large number of snapshots, and once it has a sufficient data set, the periodic checker can be set to a higher value to reduce the data transmission.
- **Event checker:** Event checkers are functions that use features to create transmission triggers based on threshold values and conditional checks.

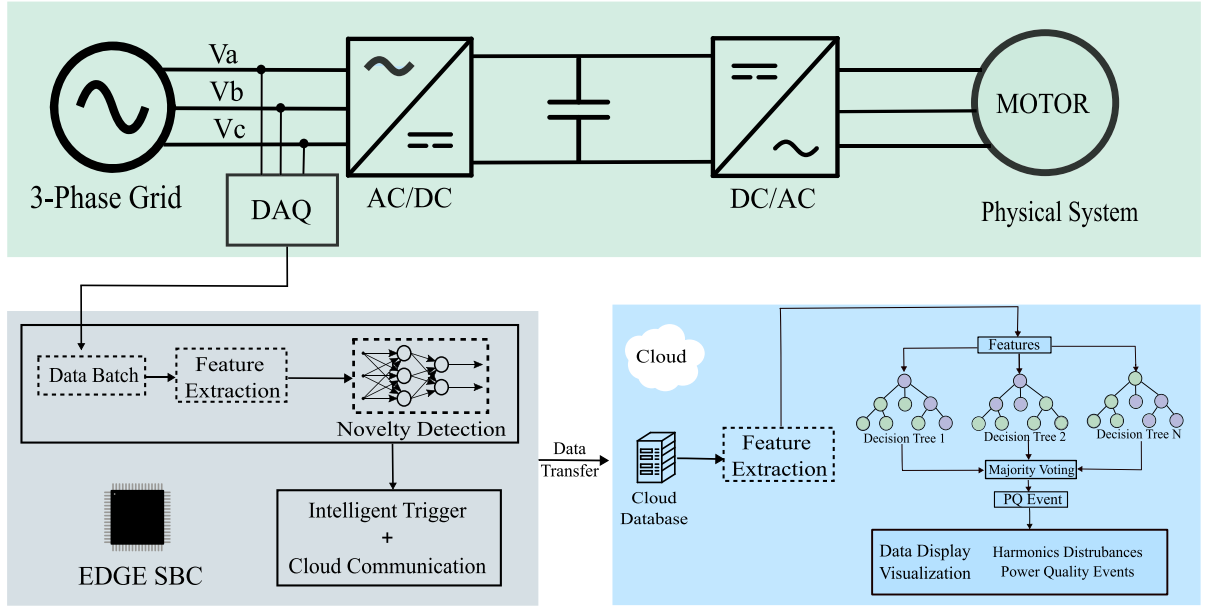


Fig. 4. Implementation details of the computing framework with a three-phase industrial motor drive, an edge device and a cloud platform.

2.2. Cloud communication

The cloud communication section of the edge software manages the data transmission with the cloud database. Initially, the snapshot data is converted into Hierarchical Data Format version 5 (HDF5) format through a designated function. The edge software attempts to send the data to the cloud database. In the current software version, HTTP protocol is supported for cloud data transfer. If the attempted data transfer fails, the edge stores the HDF5 file in local storage and re-attempts to send the data at the next snapshot data arrival. This ensures that the edge remains resilient to network disruption and data loss. The current software version lacks a latency management solution for edge–cloud communication, as the current application does not receive critical operational feedback from the cloud software.

2.2.1. Cloud functionalities

A cloud-based platform is necessary to store the raw data collected from PESs during both normal and abnormal operating conditions, as well as all relevant features and health indicators. Cloud storage includes transmitted data from multiple edge devices. This data can be used for several analyses, such as fault severity and fault progression analysis over the fleet of PESs, AD across a fleet of PESs, the creation of reliability models of the PESs and for sensitivity, credibility and trustworthy validation of the models. To facilitate analysis and access to data from edge devices, the cloud platform should also include a visualization tool. The development of specifics of the cloud framework is in progress, as the possibilities are very extensive. In the current study, the cloud is used for the storage and real-time characterization of transmitted data.

3. Application to power quality monitoring

The three-phase grid power supply is a commonly used method of providing power to manufacturing sites, mobile towers, data centers, consumer loads, and charging stations that comprise electric drives and PESs. However, fluctuations in the input voltage of the three-phase grid power supply can lead to disruptions in the power system, increased power losses, and damage to electrical equipment [20,24]. According to the IEEE Standard 519-2014, the power system harmonic limits, voltage fluctuation, and flickers that are responsible for grid fluctuations should be monitored and kept to a minimum to ensure efficient and reliable operation of the system. The importance of PQ monitoring is on the rise due to the exponential increase in distributed generation, power electronic equipment, and renewable energy sources [25]. Several standards, including IEEE 1159, IEC 61000-4-30, and EN50160, provide definitions of PQ events and indices [26]. The IEEE 1159 standard in particular provides detailed definitions of various PQ events. The objective of the developed PQ monitoring software is to detect abnormalities in the input voltage waveforms and classify the PQ event that has occurred. In this article, the focus is on the detection of the following PQ events:

- Normal voltage waveforms: As per EN 50160 [28], the recommended permissible variation in the nominal Root Mean Square (RMS) voltage is limited to 10%. Considering a 230 V Phase-Neutral nominal power supply an RMS variation of 210 V - 250 V, fundamental frequency (f) 50 Hz is considered as a normal voltage waveform.

Table 1
Mathematical equations for PQ events [26]–[27].

Category	Spectral information	Equation	Parameters/Remarks
Normal Voltage	Steady State	$V_{peak} \sin(\omega t - \phi)$	Normal RMS voltage : 210 V–250 V Phase to Neutral
Voltage Sag	Duration : 10 ms–3 s	$V_{peak}(1 - \alpha(u(t - t_1) - u(t - t_2)))\sin(\omega t - \phi)$	$0.1 \leq \alpha \leq 0.9$, u : step function t_1, t_2 : Start, End time
Voltage Swell	Duration: 10 ms–3 s	$V_{peak}(1 + \beta(u(t - t_1) - u(t - t_2)))\sin(\omega t - \phi)$	$0.1 \leq \beta \leq 0.8$ u : step function t_1, t_2 : Start, End time
Voltage Interruption	Duration: 10 ms–3 s	$V_{peak}(1 + \gamma(u(t - t_1) - u(t - t_2)))\sin(\omega t - \phi)$	$0.9 \leq \gamma \leq 1.0$ u : step function t_1, t_2 : Start, End time
Flicker	Spectrum: Upto 9 kHz Steady State	$V_{peak}(1 + \lambda \sin(\omega_f t))\sin(\omega t - \phi)$	$0.05 \leq \lambda \leq 0.1$ $f_f \in [8 \text{ Hz}, 25 \text{ Hz}]$, $\omega_f = 2\pi f_f$

- Voltage Sag, Swell, Interruption, Flickering. The details of the parameter variation are listed in Table 1.

The cloud-edge computing framework and intelligent PQ monitoring is implemented as illustrated in Fig. 4. The physical hardware used for the validation of the framework comprises an industrial motor drive with an inverter that drives the motor. The DC link capacitor is supplied with power from a three-phase grid supply through a passive AC/DC rectifier. The DAQ utilized in this setup includes three high-voltage isolated sensors with an analogue bandwidth of 100 kHz (i.e., an absolute accuracy 0.15%) and a dSPACE for sampling the data at 4 kHz. Then dSPACE sends the acquired data to an edge device, a Tinker Board 2.0 (Asus) single board computer, via the Ethernet TCP/IP protocol. The Tinker Board is equipped with a Rockchip Quad-Core RK3288-CG.W processor, 2 GB of RAM, and both WiFi and LAN communication capabilities. The edge computing framework is developed in Python3 in [21]. Some of the major Python modules utilized in the development of this framework and the PQ monitoring include Pandas, Numpy, SciPy, and sci-kit-learn. A local personal computer running the Minio-Cloud Object storage platform was used to develop the cloud system. This cloud server supports remote data storage and is compatible with the AWS S3 format. The local cloud server runs Python for additional data analysis and visualization capabilities. The cloud PC and Tinker board are connected to campus-wide WiFi service, operating with IEEE 802.11ac 5 GHz protocol. The edge software packages the snapshot into HDF5 format, and sends the h5 file to the cloud PC via Hypertext Transfer Protocol (HTTP).

The computing load for PQ monitoring is shared between the edge and the cloud. The edge receives the sampled voltage data using the Ethernet protocol. The received data are then batched into snapshots, as shown in Fig. 3. The detection of abnormal voltage waveforms is performed using an Artificial Neural Network (ANN). The ANN is a regression model that assigns a numerical value, referred to as a novelty score, to each snapshot to indicate the degree of abnormality present in it. The feature extraction, number of features, and ANN network have been optimized to require minimal computational resources. The selection of data to transmit to the cloud is based on the upper and lower thresholds of the novelty score. Any snapshot that violates the novelty score thresholds is sent to the cloud along with periodic snapshots. The received data in the cloud are further processed to extract advanced features. A Random Forest (RF), which is an ensemble of multiple Decision Trees (DT) with a majority voting scheme between DTs is used to classify the snapshot into what kind of PQ event has occurred and in which phase it has occurred.

To develop the edge and cloud algorithms for this study, a training data set was created that includes a variety of PQ events. This training data set was generated using mathematical equations as shown in Table 1. In addition, real-field data was also collected and added to the training data set. It was observed that real-field data waveforms often had a higher harmonic content than mathematically generated normal voltage waveforms. The training data set comprises three-phase line-to-line voltage waveforms (i.e., Vab, Vbc, Vca). The equations in Table 1 are for Va, Vb, and Vc. This allows the cloud-based RF classifier to identify in which phase the PQ event occurred. The developed software was tested and validated using an in-house grid emulator to generate artificial PQ events, as well as with an extensive 7-day testing period on a real three-phase grid.

4. Edge software development

In this section, the implementation of the edge software development aspect of the framework is presented. The single board computer is utilized to receive data via Ethernet communication protocol. The received data are then decoded and processed to extract sensor data for feature extraction and classification of normal and abnormal PQ snapshots.

4.1. Feature extraction

The Short-Time Fourier Transform (STFT) is a widely used method for evaluating time-varying waveform distortions in PQ analysis [29]. Because PQ disturbances are non-stationary, STFT is a useful tool for their analysis. STFT involves applying a fixed-length moving window to the signal and performing a Fourier Transform to obtain both time and frequency information. The discrete STFT for a signal $x[n]$ is given in Eq. (1), as detailed in [citeoppenheim1999discrete].

$$X(m, k) = \sum_{n=-\infty}^{\infty} x[n]w[n - m]e^{-j2\pi kn/N} \quad (1)$$

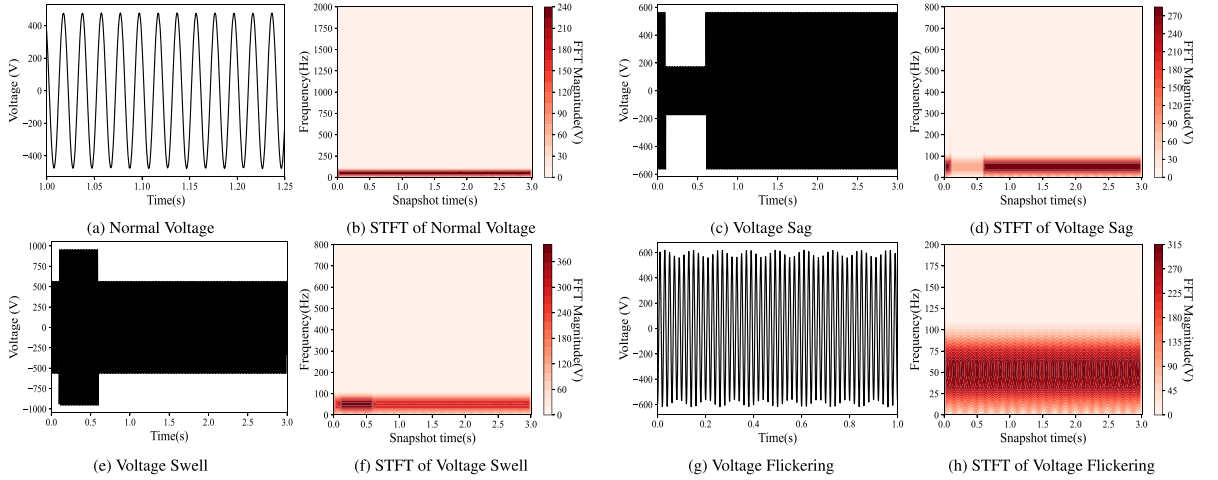


Fig. 5. STFT for various PQ disturbances. Figure a,b: STFT for normal voltage waveforms. c,d: STFT for voltage sag, a clear decrease in the coefficient magnitude is visible in Figure d. e,f: STFT for voltage swell, a clear increase in the coefficient magnitude is visible in Figure f. g,h: STFT for a flickering voltage waveform, the time variation of coefficient magnitude is visible.

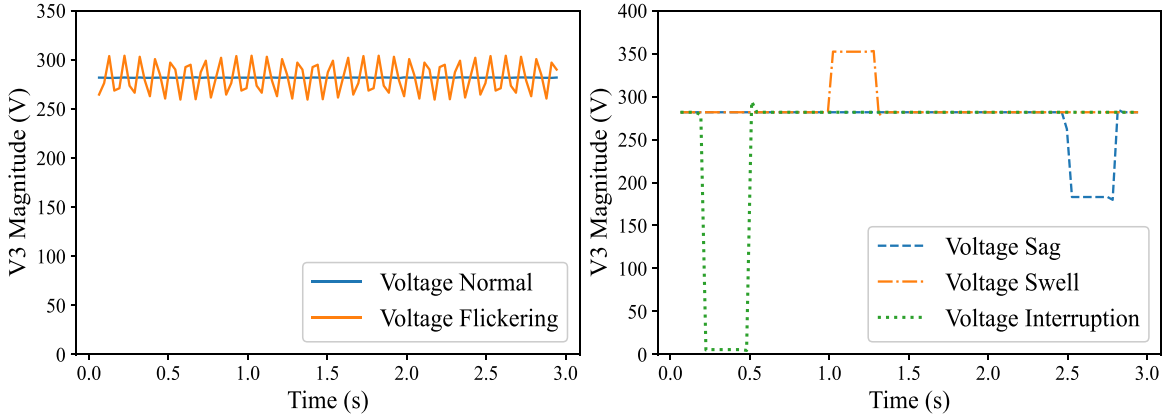


Fig. 6. STFT 3rd harmonic coefficient V3 (46.875 Hz) variation for PQ events.

where w represents the window function, m is the position of the window, and $k\omega_0$ refers to the discrete frequency, where ω_0 is the frequency resolution of the applied Fourier transform. The time and frequency resolution of the STFT is fixed, as the window length is selected in advance. The overlap between consecutive windows also impacts the time resolution of the output. STFT returns FFT coefficients for each window. Eq. (2) gives the time and frequency resolution of STFT.

$$\Delta t = \frac{N - N_o}{F_s} \quad \Delta f = \frac{F_s}{N} \quad (2)$$

where F_s is the sampling frequency of the data, N is the length of the window on which the FFT is performed, and N_o is the number of samples of overlap. A sampling frequency of 4000 kHz and snapshot length of 3s were selected, considering the constraints of the DAQ and single board computer on the sampling rate and data transfer. The STFT was applied with $N = 256$ and $N_o = 128$, resulting in a time resolution of $\Delta t = 32$ ms and a frequency resolution of $\Delta f = 15.62$ Hz. Fig. 5 shows a few examples of the STFT data extracted from the different types of training data sets.

As shown in Fig. 5, certain frequency components are particularly useful for identifying the presence of PQ events. These components are typically located near the fundamental frequency (50 Hz). The specific frequency of 46.875 Hz (3rd harmonics) was chosen to evaluate the PQ abnormalities present in the voltage snapshot. Fig. 6 illustrates the V3 (3rd harmonics component) of the STFT of voltage waveforms under different PQ conditions. A cumulative feature is also extracted, which is the maximum STFT magnitude between the 8th (125 Hz) and 128th (2000 Hz) harmonics, to evaluate the level of harmonics present in the voltage waveform.

The widespread use of ML models and AI techniques has led to the development of libraries for feature extraction. In [30] the tsfresh package in Python for extracting features from time-series data is proposed. The module provides 77 unique feature functions

Table 2
Extracted feature list [30].

No.	Feature name
1	Absolute Energy
2	Absolute Maximum
3	Absolute sum of changes
4	Binned Entropy
5	c3 Statistics for non-linear measurement [31]
6	Time series complexity [32]
7	Percentage of reoccurring data points
8	Kurtosis
9	Count above mean
10	Sample Count below mean value
11	Mean Value
12	Minimum Value
13	Skewness
14	Harmonic Content

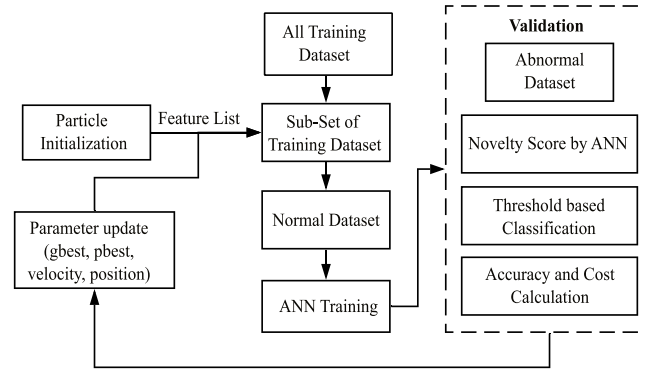


Fig. 7. Development of a Novelty Detection MLP with PSO Optimization algorithm.

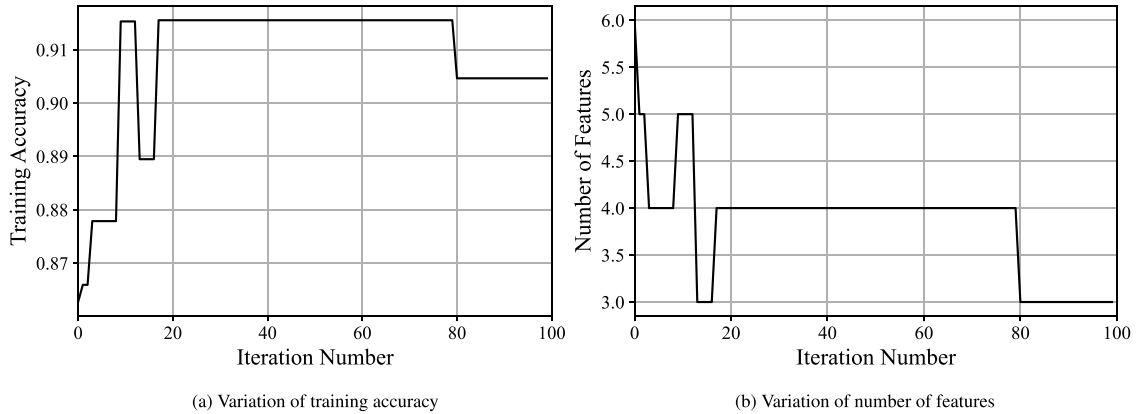


Fig. 8. Results of ANN Training and Optimization: Selection of ANN size (a, b, c) and output of the ANN with training dataset d, e .

and functions for optimized feature selection. Table 2 lists the selected features to be used in the development of the ANN and RF models.

4.2. ANN for novelty detection with particle swarm algorithm

An ML model is employed to classify grid input voltage snapshots as either normal or abnormal based on the extracted features. The grid voltage normal waveforms are clearly defined and known, so it is appropriate to develop an algorithm for detecting

abnormal grid voltages that utilize only training data of normal voltages. Novelty detection is a technique to detect anomalies in the input data set and the model is trained on a data set that does not contain any anomalous data points [33]. There are a variety of methods for implementing novelty detection, including those based on multi-layer perceptron (MLP), support vector machines (SVM), adaptive resonance theory, and radial basis function, as reviewed in [34,35].

The popularity of deep learning has led chip manufacturers like Texas Instruments, and STMicroelectronics to start supporting neural network models in low-cost embedded hardware devices. Keeping in mind the practical implications of the work for future deployment, an MLP-based approach is used for novelty detection. A specific value is assigned to the normal data set and the regression model is trained. The goal of this training is to identify an MLP that produces a specific output value for normal input test signals. If the input data is not part of the training dataset, the MLP will produce a different output. A threshold-based function is then applied to the output of the MLP to detect any abnormalities in the captured grid voltage snapshot. The available feature space consists of 13 distinct features (excluding harmonic content) for each voltage waveform. The training dataset comprises 1500 snapshots of normal voltage signals, as well as 16497 snapshots of abnormal voltage signals of various types. The performance of the MLP model is evaluated by applying a range of thresholds to the MLP output and categorizing the input as either normal or abnormal snapshots. A discrete Particle Swarm Optimization (PSO) algorithm [36,37] is employed to optimize the feature space without compromising the effectiveness of the MLP. This approach utilizes a 1×13 list of boolean values, where a value of “True” indicates that the feature is selected and “False” indicates that the feature is not selected for the particle. The cost function for the PSO to minimize is

$$cost = \alpha \times p + (1 - \alpha) \times \eta \quad (3)$$

where p is the number of features, η is the accuracy of the MLP and α is the compromise between p and η . The total overview of the total procedure is shown in Fig. 7. The output of the PSO is shown in Fig. 9. The optimized feature space has 3 features, which are item numbers 3, 7, 12 from Table 2.

The optimized feature space is employed to train and optimize the ANN. The objectives of this training optimization include minimizing the execution time, reducing the standard deviation of the output for normal voltage snapshots and maximizing the range of the output for abnormal voltage snapshots. Figs. 8(a), 8(b) 8(c) demonstrate the optimization objectives through heat maps that take into account the number of hidden layers and the number of neurons per hidden layer. After careful consideration, an ANN with 3 hidden layers and 16 neurons is selected, which resulted in a standard deviation of 0.18 on normal voltage outputs, and an execution time of 367.1 μ s. Fig. 8(d) illustrates the novelty score output for normal voltage snapshots, which ranges between 8.3 to 11.4. Additionally, Fig. 8(e) illustrates the ANN output for abnormal voltage snapshots of various types. Upon conducting successive testing, it was discovered that the novelty scores assigned by the ANN for normal voltage snapshots may be affected by an offset, which is based on the harmonic content of the voltage signal. To address this issue, a feedback loop from the cloud software was implemented to adjust the assigned thresholds on the novelty score. The edge software reads the adjusted thresholds of the novelty score from the cloud in real time.

5. Cloud software development

The development of the cloud software for this study includes the creation of a classifier for identifying abnormal voltage snapshots transmitted by edge devices, cloud storage capabilities for features and snapshots and an HTML-based visualization tool. There are a variety of methods that have been applied successfully to PQ classification problems, such as DT, ANN, rule-based expert systems and SVM [25,38]. For this study, an RF classifier is utilized. RF is an ensemble ML model, which improves classification accuracy and predictive consistency by utilizing multiple DTs and a majority voting mechanism for final classification. RF is inherently a multi-output model and it is less prone to overfitting compared to DTs, as the feature space is randomized for each DT. The complete feature set listed in Table 2 is used for the training of the RF model. Two RF models are trained on the full dataset.

- PQ event classifier: Detect the PQ of the voltage snapshot and classify the snapshot for ‘Normal’, ‘Sag’, ‘Swell’, ‘Interruption’ and ‘Flicker’ events.
- PQ phase classifier: Detect the phase in which the PQ has occurred. The classification outputs are ‘a’, ‘b’, ‘c’, ‘ab’, ‘bc’, ‘ca’, and ‘abc’.

The final classifier function is an ensemble of both models. The training of the model is optimized for the number of DTs required to achieve peak classification accuracy. Fig. 10 shows the accuracy of both models concerning the number of DTs in the model. A peak accuracy of 93.8% and 96.15% is achieved at 50 and 460 trees for the PQ event and phase classifier respectively. Table 3 shows the training validation F1-score for various classes. The F1-score is defined in Eq. (4), which is described in [39].

$$\begin{aligned} \text{Precision} &= \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalsePositive}} \\ \text{Recall} &= \frac{\text{TruePositive}}{\text{TruePositive} + \text{FalseNegative}} \\ \text{F1-score} &= 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \end{aligned} \quad (4)$$

A web-based dashboard was developed to provide a visual representation of the real-time testing and cloud data within the Python environment. The dashboard includes appropriate visualization of voltage snapshots, identified faults, and harmonic content of the grid voltages.

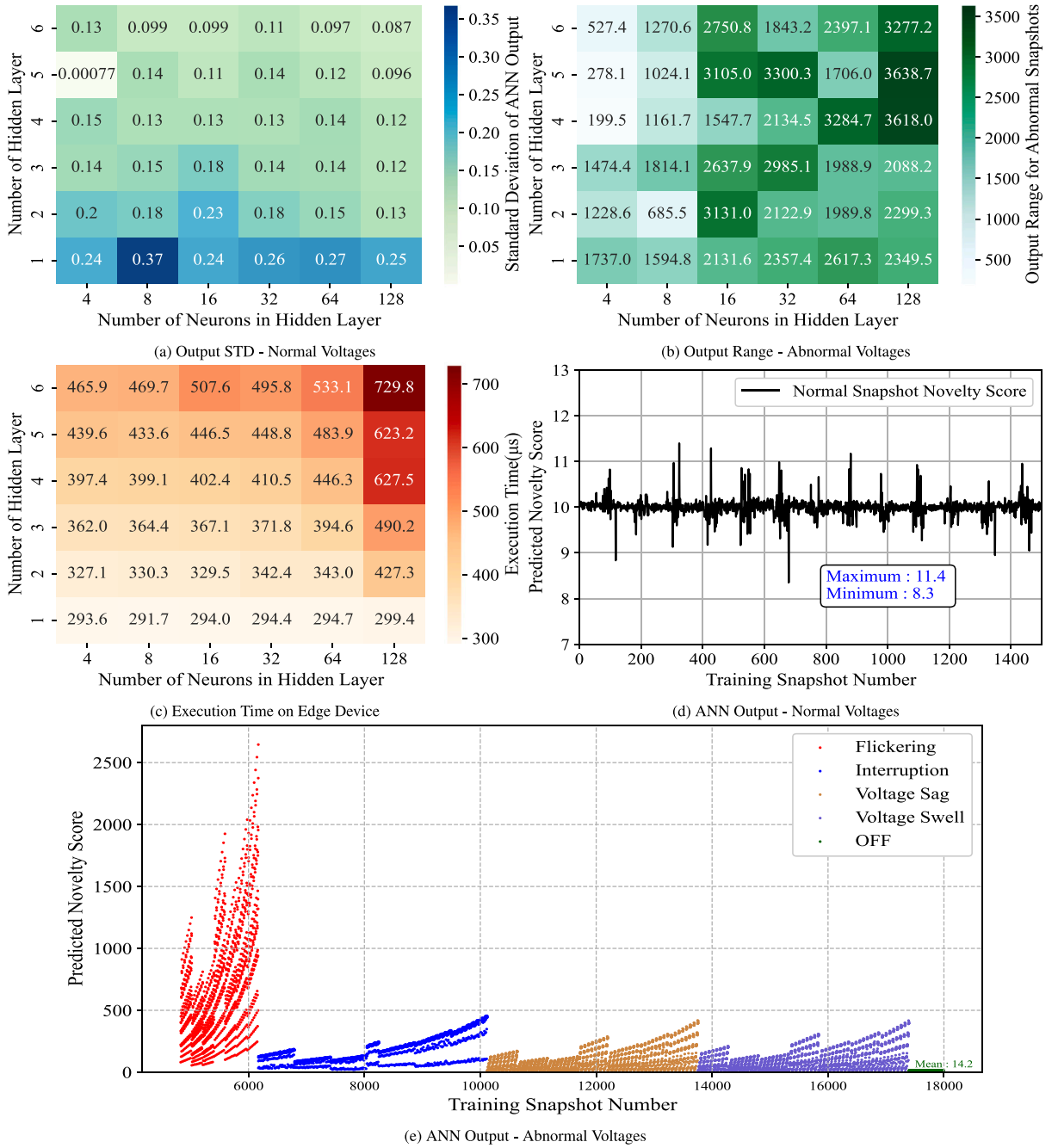


Fig. 9. PSO optimization output showing optimization of the number of features.

6. Hardware implementation, results and discussions

The hardware implementation has 3 parts to it

- **Physical System:** 1. An industrial inverter motor drive powered by the three-phase grid from a grid emulator via a passive rectifier operating at 10 kW. 2. A raw three-phase grid supply cabinet that supplies various working equipment within the laboratory.
- **Edge:** Tinker Board connected dSPACE for data acquisition and running the developed edge code.

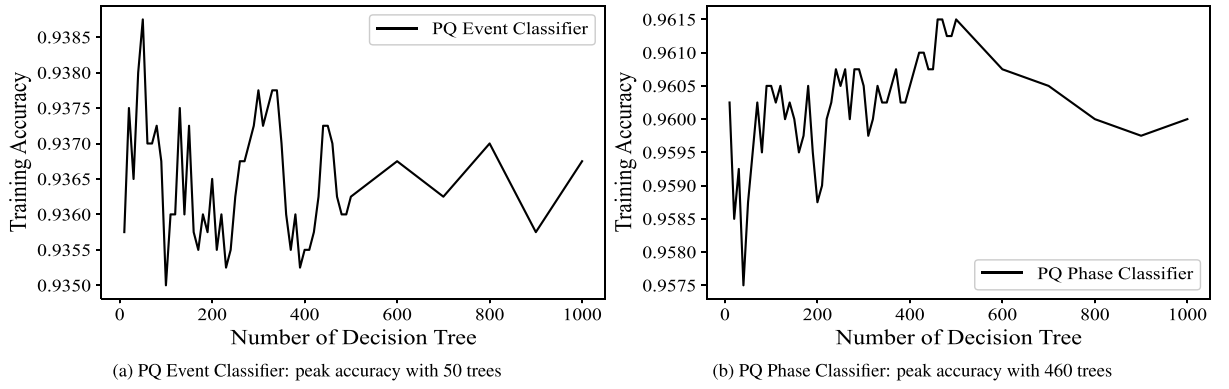


Fig. 10. Training accuracy vs number of DTs used in RF modeling.

Table 3

Training validation of the combined RF model.

Class	F1-score	Class	F1-score
Flicker_abc	1.0	Sag_c	0.86
Normal	0.9	Sag_ab	0.92
OFF	1.0	Sag_bc	0.92
Interruption_a	0.93	Sag_ca	0.96
Interruption_b	0.97	Sag_abc	0.95
Interruption_c	0.95	Swell_a	1.0
Interruption_ab	0.98	Swell_b	1.0
Interruption_bc	0.97	Swell_c	1.0
Interruption_ca	0.96	Swell_ab	1.0
Interruption_abc	0.98	Swell_bc	0.89
Sag_a	0.84	Swell_ca	0.89
Sag_b	0.88	Swell_abc	0.87

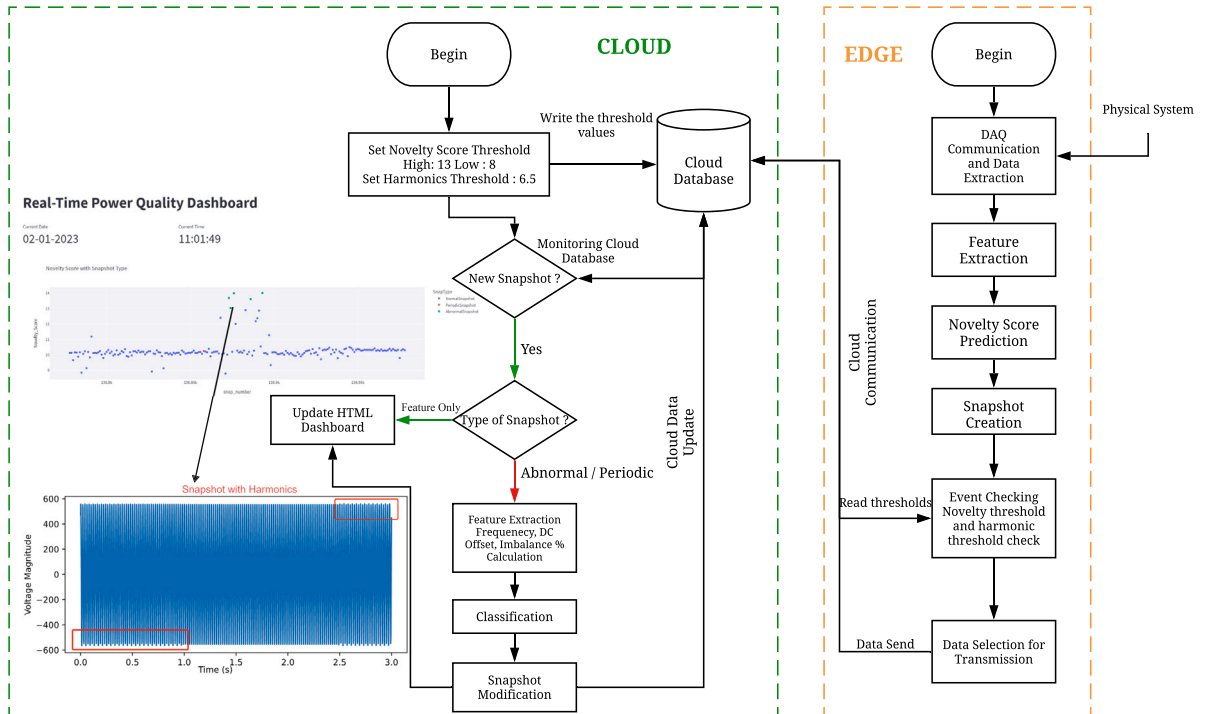


Fig. 11. Algorithm overview of edge-cloud code running the framework for PQ monitoring.

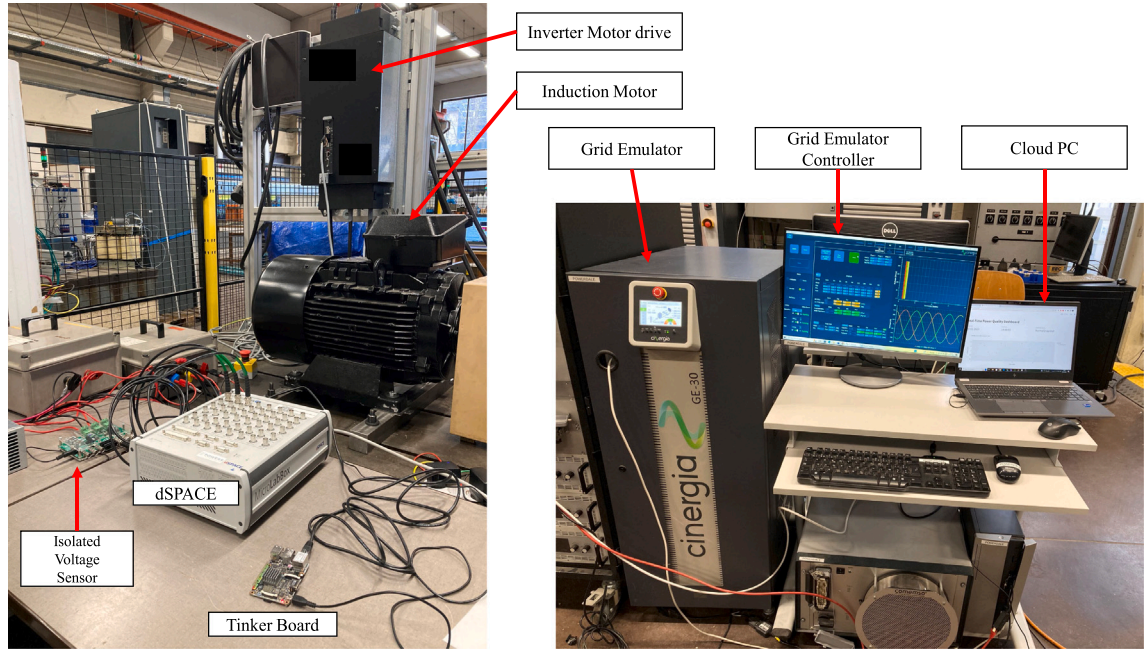


Fig. 12. Hardware setup for testing with the grid emulator.

- **Cloud:** A local PC running Minio-Cloud Storage along with Python to communicate with edge and process the incoming snapshots.

A detailed overview and flowchart of the entire edge–cloud framework is presented in Fig. 11. The cloud initiates the framework and sets the appropriate thresholds for the novelty detectors and harmonic content checkers on the edge device. The edge device processes the acquired data, extracts relevant features and executes the ANN to assign a novelty score. The voltage snapshot is then created and passed through the threshold checkers. The full data is transmitted to the cloud only when the voltage snapshot does not comply with the threshold rules, otherwise, only the features are sent. Additionally, the edge devices send a complete voltage snapshot to the cloud every 5 min periodically. The cloud also initiates a web-based HTML dashboard for visualization purposes. The new snapshots are always checked and the features are displayed on the dashboard. For the snapshots with full data, additional features are extracted and passed through an RF classifier to determine the type of voltage waveform. The classification data is also displayed on the dashboard. The cloud also includes additional functionalities to calculate the voltage imbalance percentage and the fundamental frequency of the voltage snapshot.

6.1. Performance validation of the proposed solution

The hardware setup for performance validation of the developed system is shown in Fig. 12. An industrial induction motor drive operating at 10 kW is fed three-phase grid voltage using a grid emulator. PQ events are emulated by the grid emulator, allowing control over various fault parameters. The faults were introduced in a combination of multiple phases and multiple parameters where the fault timing and fault percentage were also varied. A total of 590 distinct PQ event snapshots are generated for extensive testing. The assigned novelty score by ANN in the normal class is around 22 for this test. This is due to the absence of harmonics in the grid emulator output, which results in a pure sine-wave pattern, which differs from the typical voltage patterns observed in real grid systems. The thresholds on the edge were updated remotely from the cloud. The summary of the test results is presented in Table 4. For every PQ event, the parameters as listed in 1 are varied in the grid emulator. Table 4 provides details of each PQ event, the accuracy of event detection, the phase of occurrence, and remarks on loss of accuracy. The overall accuracy for correctly classifying PQ events is found to be 94.6%, with a 95.6% detection rate for the phase of occurrence, resulting in an overall accuracy of 91.9% for the entire system.

6.2. Testing with three-phase grid voltage

Another experimental test was carried out for a duration of 7 days on a three-phase grid supply, which provides power to the laboratory equipment. The aim of this test was twofold: to assess the performance of the developed system on an actual grid and to estimate the data transmission reduction. During the test, the entire framework remained operational for the entire 7-day 24-hour period. The results of the test are visualized in Fig. 13. Each dot in Fig. 13(a) represents a voltage snapshot and the red dots signify

Table 4

Test results of the cloud PQ classifier on an industrial inverter motor drive setup with an overall accuracy of 92%.

Power quality	Number of test snapshot	Accuracy			Novelty score (mean)	Remark on loss of accuracy
		Event	Phase	Event and phase		
Normal	55	96.3%	100.0%	96.3%	22	Some Low % voltage sag/swell (below 10%) mis-classified
Voltage Sag	226	94.2%	95.6%	91.2%	117	Some voltage sag from 85%–90% got mis-classified as interruption
Voltage Swell	161	98.1%	92.5%	92.5%	48	Mis-classification of swell phase to 'abc' in few snapshots
Voltage Interruption	94	93.5%	96.8%	93.5%	181	Mis-classification to voltage sag
Flickering	54	85.2%	100.0%	85.2%	246	Snapshots are misclassified as voltage swell, when flickers of higher percentage concerning nominal voltage are given.
Total	590	94.6%	95.6%	91.9%	–	

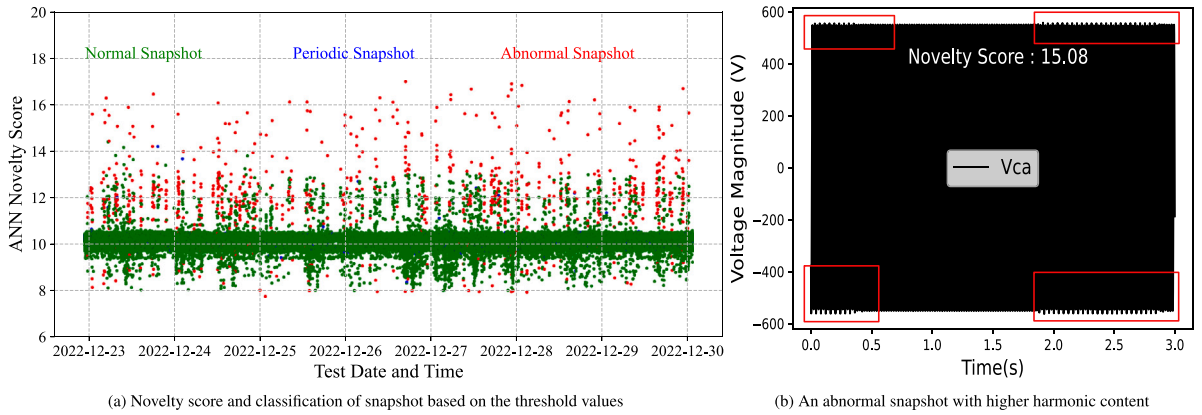


Fig. 13. Testing result with three-phase grid voltage: 7 days of continuous testing.

abnormal snapshots that were transmitted to the cloud. All the received abnormal voltage snapshots were identified as flickering and high harmonic voltage waveforms. Fig. 13(b) illustrates a typical faulty snapshot received from the edge device, with a novelty score of 15.8, displaying visible harmonic contents. The summary of the transferred data is

- Total number of received snapshots: 204513
- Number of abnormal snapshot: 972 (290 MB)
- Number of periodic snapshot: 1989 (566 MB)
- Number of feature-only snapshot: 201643 (2.8 GB)

The data transmission of 204513 snapshots with full data would amount to 56.8 GB. A 93.6% reduction in data transmission is observed. Assuming the cost of transmission on a 4G network is €2.13/GB, a cost saving of €5.9k can be achieved yearly on the data transmission cost.

6.3. Comparative analysis of the proposed solution

Monitoring of PES specifically requires electrical signals sampled at high frequency. As stated above if only a cloud-based PQ monitoring had been adapted with the same level of functionality, it would take transmission of 8 GB of data per day to the cloud. It would increase the data transmission cost significantly. In literature only-cloud based PQ monitoring is restricted to limited capability. The cloud-based PQ monitoring is only capable of storing limited features of voltage and current data, for visualization purposes [44,45]. Table 5 provides a comparison between recent works that are very similar to the work that has been carried out in the paper. The authors find the contribution of the current work towards a generalized PES monitoring IoT platform, and

Table 5
Comparative assessment of similar work.

Ref.	Context of work	Edge device used	Major computation of edge	Advantage and limitation
[40]	Edge-cloud computing for power electronics converter fleet	Rapsberry Pi	Distributed learning with deep learning models	- Implemented multiple problems in the power electronics domain with federated learning - Lack of specific edge software structure and data transmission strategy
[41]	Real-time monitoring of smart grid to prevent safety accidents and improve operational efficiency	Smart HD WiFi camera with Dell server (i7-3770)	Heuristic algorithm using simulated annealing strategy	- Introduce edge computing to smart grid application - Comparison between edge computing to cloud-only computing showing 85% reduction in detection latency and 10 times increase in frame rate - A generalized and scalable edge computing architecture is not provided to cover a wide range of applications in electrical engineering
[42]	Edge-cloud based PQ monitoring	National instruments DAQ with processing on local computer	Data processing and calculation of Power, Unbalance voltage and thresholds for PQ detection	- Detection of power quality using calculated time domain features from voltage signals and reduction of cloud data storage by selectively transmitting data with PQ events - Frequency analysis of voltage signal is not carried out, so the system is not capable of detecting frequency domain PQ events namely. flickering and high harmonic content effectively - The detailed PQ classification is missing, which is a future work the author has acknowledged.
[43]	IoT-based portable PQ monitoring device	STM32 microprocessor with peripherals for communications	Calculation of harmonic distortion, power factor, frequency, RMS voltage and current	- A portable edge device is built to process PQ data in real-time and store the data in the cloud. - The cloud lacks sampled high-frequency voltage data on the cloud and fault-specific data.
Current work	Intelligent condition monitoring of PES devices with use case as PQ monitoring of Industrial drive system	Tinker Board 2.0 (Asus)	Frequency domain analysis and classification of between normal and abnormal voltage using optimized ANN	- Implementation of an edge-cloud framework on industrial setup, with generalized application on PESs on vision. The framework is scalable to a fleet of different PES, as all the edge nodes share the same developed edge software framework and snapshot data structure. - The cloud received high-frequency voltage signal data of PQ events and periodical normal data. - Cloud further classifies the data for specific types of PQ event analysis namely, Sag, Swell, Interruption, and Flickering along with the phase in which it has occurred.

showcasing the use of ML models in the proposed architecture for solving PQ monitoring. The work is scalable in nature, as the multiple edge sites will share the same edge software and snapshot data structure.

7. Conclusion

This article has proposed an intelligent and cost-effective machine learning-based cloud-edge framework and envisions its application for next-generation PESs for CM/AD and PHM applications, in order to achieve higher efficiency, reliability and availability. This ML-based efficient cloud-edge framework has the possibility to be utilized extensively for developing data-driven physics-based digital twin and predictive/preventive health management strategies, where the data exchange model for digital twins/management strategies requires information from multiple edge nodes. The primary purpose of the framework is to improve data processing efficiency, speed, bandwidth and latency, eliminating the need to send high-volume and high-frequency data generated in the field during real-life operation to the cloud platform, and rather perform (pre-)processing in the field. Additionally, this framework provides proximity to data at its source and delivers strong business benefits such as faster insights, improved response times and better bandwidth availability.

The practical application and feasibility of the framework have been demonstrated through monitoring the PQ in a three-phase grid for an industrial drive setup. The framework has been further analyzed from the edge computing perspective, including deploying ML models on edge devices and intelligent communication with the cloud. The cloud code also features the deployment of larger ML models for post-processing on received raw data. Additionally, the case study has demonstrated the application of one-class classification of voltage waveforms using STFT and an optimized ANN. The ability of a novelty detector trained on a limited dataset to detect abnormalities in untrained data has also been highlighted. The achieved data transmission reduction is 94% with a cost-saving of up to €5.9k per year on data transmission. The cloud employs multiple RF classifiers to detect relevant PQ events with 91.2% accuracy. A week-long test also reveals the common occurrence of harmonic distortions and flickering events on a three-phase grid supply.

This paper showcases a real-life implementation of the IIoT edge-cloud framework on industrial setup, edge device and cloud. The data transmission reduction ratio will vary across applications, preventing a direct comparison with existing literature. The application of the framework to the PQ problem uniquely uses a computationally efficient ANN to identify PQ events. While the accuracy of the cloud-based PQ classifier could be enhanced, the study underscores the seamless integration of ML models into IoT architectures. The main limitation of work is that the IIoT platform is not fully ready to be deployed in the field. The edge software is not optimized for IoT devices, to run in a container. Also, the developed PQ monitoring needs to be deployed for a fleet of industrial setups to monitor multiple sites. This opens doors for future research works such as

1. Practical implementation of data acquisition hardware to be used in PESs to sample signals at high frequency. For example sampling of DC link voltage of an inverter for CM application
2. Deployment of the edge software to multiple edge devices and transferring learning between multiple edges
3. Containerization of edge software for deployment and further leveraging techniques such as code profiling, algorithmic optimizations, and utilization of optimized libraries to enhance execution efficiency
4. Development of cloud software to accommodate digital twin models and fleet management services

Future work would address these possibilities with the framework and develop the framework to address the requirements arising from the applications.

CRedit authorship contribution statement

Sachin Kumar Bhoi: Writing – original draft, Visualization, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Sajib Chakraborty:** Writing – review & editing, Resources, Project administration, Methodology, Investigation, Funding acquisition, Conceptualization. **Boud Verbrugge:** Writing – review & editing, Validation, Software, Resources, Methodology, Investigation, Formal analysis, Conceptualization. **Stijn Helsen:** Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Steven Robyns:** Visualization, Validation, Software, Methodology, Data curation, Conceptualization. **Mohamed El Baghdadi:** Supervision. **Omar Hegazy:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The data that has been used is confidential.

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